
RAJSHAHI UNIVERSITY OF ENGINEERING &
TECHNOLOGY

Department of Computer Science & Engineering

Numerical Methods Laboratory Manual

Solution of Algebraic & Transcendental Equations

Chapter 2 · Root-Finding Methods

Prepared by: Md. Azmain Yakin Srizon
Assistant Professor
Dept. of Computer Science & Engineering, RUET

Topics: Bisection, False Position, Iteration, Aitken's Δ^2 ,
Newton–Raphson, Generalized Newton, Secant,
Ramanujan's Method, and Systems of Nonlinear Equations

Reference: S.S. Sastry, *Introductory Methods of
Numerical Analysis*, 5th ed., PHI Learning

Contents

1	Introduction and How to Use This Manual	2
2	Lab Report: Format and Submission Guidelines	2
2.1	Report structure	2
2.2	Formatting rules	3
2.3	Submission rules	3
3	Experiment 1: Bisection Method	4
3.1	Theory	4
3.2	Worked Example (full input and output)	4
4	Experiment 2: Method of False Position (Regula Falsi)	6
4.1	Theory	6
4.2	Worked Example (full input and output)	6
5	Experiment 3: Fixed-Point Iteration Method	8
5.1	Theory	8
5.2	Worked Example (full input and output)	8
6	Experiment 4: Aitken's Δ^2 Acceleration	10
6.1	Theory	10
6.2	Worked Example (full input and output)	10
7	Experiment 5: Newton–Raphson Method	12
7.1	Theory	12
7.2	Worked Example (full input and output)	12
8	Experiment 6: Generalized Newton's Method (Multiple Roots)	14
8.1	Theory	14
8.2	Worked Example (full input and output)	14
9	Experiment 7: Secant Method	16
9.1	Theory	16
9.2	Worked Example (full input and output)	16
10	Experiment 8: Ramanujan's Method (Smallest Root)	18
10.1	Theory	18
10.2	Worked Example (full input and output)	18
11	Experiment 9: Iteration Method for Systems of Nonlinear Equations	20
11.1	Theory	20
11.2	Worked Example (full input and output)	20
12	Experiment 10: Newton–Raphson Method for Systems	22
12.1	Theory	22
12.2	Worked Example (full input and output)	22
	Appendix: Quick Comparison of the Methods	24

1 Introduction and How to Use This Manual

This manual accompanies the lecture material on **Chapter 2: Solution of Algebraic and Transcendental Equations**. Each experiment treats one *root-finding algorithm* and is organised uniformly:

- **Objective** — what the experiment teaches.
- **Theory** — a short statement of the method and its convergence condition.
- **Algorithm** — complete *pseudocode* you must implement.
- **Worked Example** — a fully solved example showing the *exact input* and the *complete output* (iteration table and final root).
- **Exercise** — at least ten problems to be solved and reported.

Convention. Throughout, ε denotes the tolerance, N the maximum number of iterations, and “ $x_n \rightarrow \xi$ ” means the iterates converge to the root ξ . Unless stated otherwise, report answers correct to **four decimal places** and tabulate *every* iteration.

A general-purpose driver (in C/C++/Python/MATLAB) is recommended: read the function f (and f' where required), the starting data and ε , then loop until the stopping test is met, printing one row of the iteration table each pass.

2 Lab Report: Format and Submission Guidelines

Every submitted report **must** follow the structure and rules below.

2.1 Report structure

1. **Cover page** — Course title and code, Experiment number and name, your full name, roll/registration number, section, group, date of performance, date of submission.
2. **Objective** — one or two lines.
3. **Theory** — the method, its formula and convergence condition (in your own words; 4–6 lines).
4. **Algorithm / Flowchart** — the pseudocode (or a flowchart) you implemented.
5. **Source code** — complete, commented program listing.
6. **Sample input** — the equation, starting values and tolerance you used.
7. **Output / Results** — the *full iteration table* and the final root. Paste the program output *and* present a neat hand-formatted table.
8. **Discussion** — number of iterations, observed order of convergence, effect of the starting value / tolerance, and any difficulty faced.
9. **Conclusion** — 2–3 lines.

2.2 Formatting rules

- Paper size **A4**; margins **2.5 cm** on all sides.
- Body font **12 pt** serif (Times/Latin Modern); line spacing **1.5**; justified text.
- Headings bold; code in a **monospaced** font (e.g. Consolas/Courier), 10–11 pt, with line numbers.
- Every table and figure must be **numbered and captioned**. Tabulate iteration data with aligned decimal points.
- Number all equations you reference.

2.3 Submission rules

- Submit a single **PDF** named `Roll_ExpNo_MethodName.pdf` — for example, `2303001_Exp05_Newton-Raphson.pdf`.
- Submit the **source files** separately in a folder/zip with the same base name.
- Deadline: **within two week** of performing the experiment unless announced otherwise. Late reports lose marks as per course policy.
- Work must be **your own**. Copied reports/code receive zero; both provider and copier are penalised. You may discuss ideas but must write and code independently.

Marking (typical). Algorithm & code correctness 40%, results & iteration tables 25%, discussion 20%, formatting & neatness 10%, viva/demo 5%.

3 Experiment 1: Bisection Method

Objective

To locate a real root of a continuous function $f(x)$ on an interval $[a, b]$ with $f(a)f(b) < 0$ by repeated halving, and to study its guaranteed (but linear) convergence.

3.1 Theory

If f is continuous on $[a, b]$ and $f(a)f(b) < 0$, then by the Intermediate Value Theorem a root lies in (a, b) . The midpoint $x = \frac{a+b}{2}$ is taken as the estimate; the half that still shows a sign change is retained. After n steps the root is trapped in an interval of width $\frac{b-a}{2^n}$, so

$$|x_n - \xi| \leq \frac{b-a}{2^n}, \quad n \geq \frac{\log((b-a)/\varepsilon)}{\log 2}.$$

Algorithm 1 Bisection Method

Input: f , a, b with $f(a)f(b) < 0$, tolerance ε , max iterations N

Output: approximate root x

```

1: for  $n = 1$  to  $N$  do
2:    $x \leftarrow (a + b)/2$ 
3:   if  $f(x) = 0$  or  $(b - a)/2 < \varepsilon$  then
4:     return  $x$ 
5:   end if
6:   if  $f(a) \cdot f(x) < 0$  then
7:      $b \leftarrow x$                                 ▷ root in  $(a, x)$ 
8:   else
9:      $a \leftarrow x$                                 ▷ root in  $(x, b)$ 
10:  end if
11: end for
12: return  $x$ 

```

3.2 Worked Example (full input and output)

Example 1.1 — $f(x) = x^3 - x - 1 = 0$

Input. $f(x) = x^3 - x - 1$; bracket $[a, b] = [1, 1.5]$ since $f(1) = -1 < 0$ and $f(1.5) = +0.875 > 0$; tolerance $\varepsilon = 10^{-3}$.

Output.

n	a	b	$x_n = \frac{a+b}{2}$	$f(x_n)$	new interval
1	1.0000	1.5000	1.250000	-0.296875	[1.25, 1.5]
2	1.2500	1.5000	1.375000	+0.224609	[1.25, 1.375]
3	1.2500	1.3750	1.312500	-0.051514	[1.3125, 1.375]
4	1.3125	1.3750	1.343750	+0.082611	[1.3125, 1.34375]
5	1.3125	1.3438	1.328125	+0.014576	[1.3125, 1.328125]
6	1.3125	1.3281	1.320313	-0.018711	[1.320313, 1.328125]

Continuing, the iterates converge to the root $\xi \approx 1.3247$ (the only real root of $x^3 - x - 1$).

Example 1.2 — $f(x) = x^3 - 2x - 5 = 0$

Input. $[a, b] = [2, 3]$ ($f(2) = -1$, $f(3) = +16$), $\varepsilon = 10^{-3}$.

Output. Successive midpoints $x_1 = 2.5$, $x_2 = 2.25$, $x_3 = 2.125$, $x_4 = 2.0625$, $x_5 = 2.09375$, $x_6 = 2.10938, \dots$, with $|x_{11} - x_{10}| = 0.0005$ giving $\boxed{\xi \approx 2.094}$ (correct to three decimal places).

Exercise 1 (Bisection)

Locate a suitable bracket $[a, b]$ (if not given) and find the indicated root, tabulating every iteration. Report to 4 decimal places unless stated.

1. $x^3 - x - 1 = 0$ in $[1, 2]$.
2. $x^3 - 2x - 5 = 0$ in $[2, 3]$ (3 d.p.).
3. $x^3 + x^2 + x + 7 = 0$ in $[-3, -2]$ (3 d.p.).
4. $x^3 - 4x - 9 = 0$ in $[2, 3]$.
5. $x^4 - x - 10 = 0$ in $[1, 2]$.
6. $x - \cos x = 0$ in $[0, 1]$.
7. $xe^x - 1 = 0$ in $[0, 1]$.
8. $3x - \cos x - 1 = 0$ in $[0, 1]$.
9. $e^x - 3x = 0$: find *both* roots, in $[0, 1]$ and in $[1, 2]$.
10. $x \sin x - 1 = 0$ in $[1, 2]$.
11. $\ln x - x + 2 = 0$ in $[3, 4]$.
12. **(Application)** A wooden sphere of radius 1 m and specific gravity 0.6 floats; its submerged depth h satisfies $h^3 - 3h^2 + 2.4 = 0$. Find h in $[1, 2]$ (3 d.p.).

4 Experiment 2: Method of False Position (Regula Falsi)

Objective

To find a real root of $f(x) = 0$ by joining $(a, f(a))$ and $(b, f(b))$ with a chord and taking its x -intercept as the next estimate, and to compare its speed with bisection.

4.1 Theory

Starting from a bracket with $f(a)f(b) < 0$, the chord meets the x -axis at

$$x = \frac{a f(b) - b f(a)}{f(b) - f(a)}.$$

The sub-interval that preserves the sign change is kept and the process is repeated. It uses the *magnitude* of f (not just its sign), so it usually converges faster than bisection, though one endpoint may stay fixed.

Algorithm 2 Method of False Position

Input: f , a, b with $f(a)f(b) < 0$, tolerance ε , max iterations N

Output: approximate root x

```

1: for  $n = 1$  to  $N$  do
2:    $x \leftarrow \frac{a f(b) - b f(a)}{f(b) - f(a)}$ 
3:   if  $|f(x)| < \varepsilon$  then
4:     return  $x$ 
5:   end if
6:   if  $f(a) \cdot f(x) < 0$  then
7:      $b \leftarrow x$ 
8:   else
9:      $a \leftarrow x$ 
10:  end if
11: end for
12: return  $x$ 
```

4.2 Worked Example (full input and output)

Example 2.1 — $f(x) = x^3 - 2x - 5 = 0$

Input. $[a, b] = [2, 3]$, $f(2) = -1$, $f(3) = +16$.

Output.

n	a	b	x_n ($f(x_n)$)
1	2.00000	3	2.05882 (−0.39080)
2	2.05882	3	2.08126 (−0.14720)
3	2.08126	3	2.08964 (−0.05468)
4	2.08964	3	2.09274 (−0.02000)
5	2.09274	3	2.09388 (−0.00745)

The right endpoint $b = 3$ stays fixed; the iterates approach $\boxed{\xi \approx 2.0945}$ (5 significant figures).

Example 2.2 — $2x - \log_{10} x - 7 = 0$

Input. $[a, b] = [3, 4]$, $f(3) = -1.4771$, $f(4) = +0.3979$.

Output. $x_1 = 3.7878$ ($f = -0.002787$), then $x_2 = 3.7893$ ($f = +0.000041$) $\Rightarrow \boxed{\xi \approx 3.789}$ (3 d.p.) in just two steps.

Exercise 2 (False Position)

1. $x^3 - 2x - 5 = 0$ in $[2, 3]$.
2. $x^{2.2} - 69 = 0$ in $[5, 8]$ (i.e. solve $x^{2.2} = 69$).
3. $2x - \log_{10} x - 7 = 0$ in $[3, 4]$.
4. $4e^{-x} \sin x - 1 = 0$ in $[0, 0.5]$.
5. $x^3 - x - 1 = 0$ in $[1, 2]$.
6. $\cos x - xe^x = 0$ in $[0, 1]$.
7. $x \log_{10} x - 1.2 = 0$ in $[2, 3]$.
8. $x^3 + x^2 - 3x - 3 = 0$ in $[1, 2]$.
9. $\ln x - x + 2 = 0$ in $[3, 4]$.
10. $x - e^{-x} = 0$ in $[0, 1]$.
11. $e^{-x} - \sin x = 0$ in $[0, 1]$.
12. **(Application)** A 10 m cable hangs between poles 8 m apart; its catenary parameter c satisfies $c \sinh(4/c) - 5 = 0$. Find c in $[3, 4]$.

5 Experiment 3: Fixed-Point Iteration Method

Objective

To solve $f(x) = 0$ by rewriting it as $x = \varphi(x)$ and iterating $x_{n+1} = \varphi(x_n)$, and to verify the convergence condition $|\varphi'(x)| < 1$.

5.1 Theory

Cast $f(x) = 0$ as $x = \varphi(x)$. Starting from x_0 , generate $x_{n+1} = \varphi(x_n)$. If $|\varphi'(x)| \leq k < 1$ in a neighbourhood of the root ξ , the iteration converges *linearly*, and a practical stopping test is $|x_n - x_{n-1}| \leq \frac{1-k}{k}\varepsilon$. The *choice* of φ decides whether (and how fast) it converges.

Algorithm 3 Fixed-Point Iteration

Input: iteration function φ (with $|\varphi'| < 1$ near the root), x_0 , ε , N

Output: approximate root x_1

```

1: for  $n = 1$  to  $N$  do
2:    $x_1 \leftarrow \varphi(x_0)$ 
3:   if  $|x_1 - x_0| < \varepsilon$  then
4:     return  $x_1$ 
5:   end if
6:    $x_0 \leftarrow x_1$ 
7: end for
8: return  $x_1$ 

```

5.2 Worked Example (full input and output)

Example 3.1 — $x^3 = 1 - x^2$ on $[0, 1]$

Input. Rearranged as $x = \varphi(x) = \frac{1}{\sqrt{x+1}}$; $\varphi'(x) = -\frac{1}{2}(x+1)^{-3/2}$, $\max |\varphi'| < 0.2$; $x_0 = 0.75$, $\varepsilon = 10^{-4}$.

Output.

n	x_n	$x_{n+1} = 1/\sqrt{x_n+1}$	$ x_{n+1} - x_n $
0	0.75000	0.75593	0.00593
1	0.75593	0.75465	0.00128
2	0.75465	0.75493	0.00028

$|x_3 - x_2| = 0.00028 < \frac{1-0.2}{0.2}\varepsilon = 4 \times 10^{-4} \Rightarrow \boxed{\xi \approx 0.7549}$ (4 d.p.).

Example 3.2 — $xe^x = 1$ (a slowly converging case)

Input. $x = \varphi(x) = e^{-x}$, $x_0 = 0.5$ (here $|\varphi'(\xi)| = e^{-0.567} \approx 0.57$, close to 1).

Output. $x_1 = 0.60653$, $x_2 = 0.54524$, $x_3 = 0.57970$, $x_4 = 0.56007, \dots$ The values oscillate and need **19** iterations to settle at $\boxed{\xi \approx 0.5671}$.

Exercise 3 (Fixed-Point Iteration)

For each, choose a rearrangement $x = \varphi(x)$ satisfying $|\varphi'| < 1$, state φ , and iterate from the given x_0 .

1. $x^3 + x^2 - 1 = 0$, $\varphi(x) = 1/\sqrt{1+x}$, $x_0 = 0.75$.
2. $2x - 3 = \cos x$, $\varphi(x) = \frac{1}{2}(3 + \cos x)$, $x_0 = 1.5$.
3. $xe^x = 1$, $\varphi(x) = e^{-x}$, $x_0 = 0.5$.
4. $\sin x = 10(x - 1)$, $\varphi(x) = 1 + \frac{\sin x}{10}$, $x_0 = 1$.
5. $x = \cos x$, $\varphi(x) = \cos x$, $x_0 = 0.5$.
6. $x^2 - x - 3 = 0$, $\varphi(x) = \sqrt{x+3}$, $x_0 = 2$.
7. $x^2 - 2x - 5 = 0$, $\varphi(x) = \sqrt{2x+5}$, $x_0 = 3$.
8. $x^3 + 4x^2 - 10 = 0$, $\varphi(x) = \frac{1}{2}\sqrt{10 - x^3}$, $x_0 = 1.5$.
9. $x^3 - 3x + 1 = 0$ (root in $[0, 1]$): find a suitable φ , $x_0 = 0.5$.
10. $e^{-x} = x$, $\varphi(x) = e^{-x}$, $x_0 = 0.5$.
11. $\tan x = x$ near 4.5: find a suitable φ .
12. For each problem above, determine $k = \max |\varphi'|$ and predict the number of iterations needed for $\varepsilon = 10^{-4}$.

6 Experiment 4: Aitken's Δ^2 Acceleration

Objective

To accelerate a linearly-convergent fixed-point iteration using Aitken's Δ^2 process, and to compare the number of iterations with the un-accelerated method.

6.1 Theory

Given three successive iterates x_{i-1}, x_i, x_{i+1} of a linearly convergent sequence, a far better estimate of the root is

$$\hat{x} = x_{i+1} - \frac{(\Delta x_i)^2}{\Delta^2 x_{i-1}} = x_{i+1} - \frac{(x_{i+1} - x_i)^2}{x_{i+1} - 2x_i + x_{i-1}}.$$

Algorithm 4 Aitken's Δ^2 Acceleration

Input: iteration function φ , x_0 , ε , N

Output: accelerated root $\hat{x}$

```

1: for  $n = 1$  to  $N$  do
2:    $x_1 \leftarrow \varphi(x_0); \quad x_2 \leftarrow \varphi(x_1)$ 
3:    $\Delta^2 \leftarrow x_2 - 2x_1 + x_0$ 
4:    $\hat{x} \leftarrow x_2 - (x_2 - x_1)^2 / \Delta^2$ 
5:   if  $|\hat{x} - x_0| < \varepsilon$  then
6:     return  $\hat{x}$ 
7:   end if
8:    $x_0 \leftarrow \hat{x}$  ▷ restart from the accelerated value
9: end for
10: return  $\hat{x}$ 

```

6.2 Worked Example (full input and output)

Example 4.1 — accelerate $x = \frac{1}{2}(3 + \cos x)$

Input. Three plain iterates $x_1 = 1.500$, $x_2 = 1.535$, $x_3 = 1.518$.

Output. $\Delta x_1 = +0.035$, $\Delta x_2 = -0.017$, $\Delta^2 x_1 = -0.052$, so

$$\hat{x} = 1.518 - \frac{(-0.017)^2}{-0.052} = 1.518 + 0.006 = \boxed{1.524}.$$

This matches the value plain iteration reaches only after *six* steps.

Exercise 4 (Aitken's Δ^2)

Apply Aitken's process to the iterations of Experiment 3 and compare the iteration counts.

1. $\varphi(x) = \frac{1}{2}(3 + \cos x)$, $x_0 = 1.5$.
2. $\varphi(x) = e^{-x}$, $x_0 = 0.5$ (accelerate the slow case of Ex. 3.2).
3. $\varphi(x) = \cos x$, $x_0 = 0.5$.
4. $\varphi(x) = 1/\sqrt{1+x}$, $x_0 = 0.75$.
5. $\varphi(x) = \sqrt{x+3}$, $x_0 = 2$.
6. $\varphi(x) = \frac{1}{2}\sqrt{10-x^3}$, $x_0 = 1.5$.
7. $\varphi(x) = 1 + \frac{\sin x}{10}$, $x_0 = 1$.
8. $\varphi(x) = (2 - x^2)^{1/3}$ for $x^3 + x^2 - 2 = 0$, $x_0 = 0.8$.
9. Given the iterates 0.6, 0.69, 0.737, perform one Aitken step.
10. Given the iterates 1.0, 0.9, 0.832, perform one Aitken step.
11. For Ex. 3.3, tabulate the plain iterates *and* the Aitken estimates side by side; report the speed-up factor.

7 Experiment 5: Newton–Raphson Method

Objective

To solve $f(x) = 0$ using the tangent-line formula $x_{n+1} = x_n - f(x_n)/f'(x_n)$ and to observe its quadratic convergence.

7.1 Theory

Expanding $f(x_0 + h) = 0$ by Taylor’s series and neglecting second-order terms gives $h = -f(x_0)/f'(x_0)$, hence

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

If x_0 is sufficiently close to a simple root ξ and $f'(\xi) \neq 0$, the error satisfies $\varepsilon_{n+1} \approx \frac{1}{2}\varepsilon_n^2 f''(\xi)/f'(\xi)$ — *second-order (quadratic) convergence*.

Algorithm 5 Newton–Raphson Method

Input: $f, f', x_0, \varepsilon, N$

Output: approximate root x_1

```

1: for  $n = 1$  to  $N$  do
2:   if  $f'(x_0) = 0$  then
3:     stop                                ▷ horizontal tangent — choose another  $x_0$ 
4:   end if
5:    $x_1 \leftarrow x_0 - f(x_0)/f'(x_0)$ 
6:   if  $|x_1 - x_0| < \varepsilon$  then
7:     return  $x_1$ 
8:   end if
9:    $x_0 \leftarrow x_1$ 
10: end for
11: return  $x_1$ 

```

7.2 Worked Example (full input and output)

Example 5.1 — $f(x) = x^3 - 2x - 5 = 0$

Input. $f'(x) = 3x^2 - 2$; $x_0 = 2$; $\varepsilon = 10^{-5}$.

Output.

n	x_n	$f(x_n)$	$f'(x_n)$
0	2.000000	−1.000000	10.0000
1	2.100000	+0.061000	11.2300
2	2.094568	+0.000186	11.1616
3	2.094551	≈ 0	—

$\xi \approx 2.0946$ in only **3 steps** (cf. 11 for bisection): the error roughly squares each iteration.

Example 5.2 — $f(x) = x \sin x + \cos x = 0$ **Input.** $f'(x) = x \cos x$; $x_0 = \pi = 3.1416$.**Output.**

n	x_n	$f(x_n)$	x_{n+1}
0	3.1416	-1.0000	2.8233
1	2.8233	-0.0662	2.7986
2	2.7986	-0.0006	2.7984
3	2.7984	0.0000	2.7984

$$\boxed{\xi \approx 2.7984}.$$

Exercise 5 (Newton–Raphson)Give $f'(x)$, then iterate from the stated x_0 .

1. $x^3 - 2x - 5 = 0$, $x_0 = 2$.
2. $x \sin x + \cos x = 0$, $x_0 = \pi$.
3. $xe^x - 1 = 0$, $x_0 = 1$.
4. $\sin x = \frac{x}{2}$, $x_0 = \frac{\pi}{2}$ (root in $[\frac{\pi}{2}, \pi]$).
5. $4e^{-x} \sin x - 1 = 0$, $x_0 = 0.2$.
6. $x^3 - x - 1 = 0$, $x_0 = 1.5$.
7. $\cos x - xe^x = 0$, $x_0 = 0.5$.
8. $x^4 - x - 10 = 0$, $x_0 = 2$.
9. $e^x - 3x = 0$: find both roots, $x_0 = 0.5$ and $x_0 = 1.5$.
10. $x - \tan x = 0$: smallest positive root, $x_0 = 4.5$.
11. Compute $\sqrt{24.5}$ via $x^2 - 24.5 = 0$, $x_0 = 5$.
12. Compute $\sqrt[3]{17}$ via $x^3 - 17 = 0$, $x_0 = 2.5$.

8 Experiment 6: Generalized Newton's Method (Multiple Roots)

Objective

To restore fast convergence at a root of *multiplicity* p using the generalized Newton formula $x_{n+1} = x_n - p f(x_n)/f'(x_n)$.

8.1 Theory

At a simple root plain Newton is quadratic, but at a root of multiplicity p it degrades to *linear*. If the multiplicity p is known, the *generalized Newton's formula*

$$x_{n+1} = x_n - p \frac{f(x_n)}{f'(x_n)}$$

recovers quadratic convergence. (For $p = 1$ it reduces to ordinary Newton.) A quick check that a multiple root is near x_0 : the values $x_0 - p \frac{f}{f'}$, $x_0 - (p-1) \frac{f'}{f''}$, $\dots$ agree.

Algorithm 6 Generalized Newton's Method

Input: f , f' , multiplicity p , x_0 , ε , N

Output: multiple root x_1

```

1: for  $n = 1$  to  $N$  do
2:    $x_1 \leftarrow x_0 - p \cdot f(x_0)/f'(x_0)$ 
3:   if  $|x_1 - x_0| < \varepsilon$  then
4:     return  $x_1$ 
5:   end if
6:    $x_0 \leftarrow x_1$ 
7: end for
8: return  $x_1$ 
```

8.2 Worked Example (full input and output)

Example 6.1 — $f(x) = x^3 - x^2 - x + 1 = 0$ (double root at $x = 1$)

Input. $f = (x-1)^2(x+1)$, so $p = 2$; $f' = 3x^2 - 2x - 1$, $f'' = 6x - 2$; $x_0 = 0.8$.

Output. At $x_0 = 0.8$: $f = 0.072$, $f' = -0.68$, $f'' = 2.8$, and

$$x_0 - 2 \frac{f}{f'} = 0.8 - 2 \frac{0.072}{-0.68} = 1.012, \quad x_0 - \frac{f'}{f''} = 0.8 - \frac{-0.68}{2.8} = 1.043,$$

both ≈ 1 , confirming a double root near unity. Refining with $x_1 = 1.01$:

$$x_2 = 1.01 - 2 \frac{f(1.01)}{f'(1.01)} = 1.01 - 0.0099 = \boxed{1.0001}.$$

Plain Newton ($p = 1$) only crawls: 0.91, 0.96, $\dots$

Exercise 6 (Generalized Newton)

Each polynomial has a repeated root. State the root and its multiplicity p , then apply the generalized formula.

1. $x^3 - x^2 - x + 1 = 0$ $[(x - 1)^2(x + 1)]$, $x_0 = 0.8$.
2. $x^3 - 3x + 2 = 0$ $[(x - 1)^2(x + 2)]$, $x_0 = 1.2$.
3. $x^3 - 5x^2 + 8x - 4 = 0$ $[(x - 1)(x - 2)^2]$, $x_0 = 1.8$.
4. $x^3 - x^2 - 8x + 12 = 0$ $[(x - 2)^2(x + 3)]$, $x_0 = 1.8$.
5. $x^3 + 3x^2 + 3x + 1 = 0$ $[(x + 1)^3]$, $x_0 = -0.7$.
6. $x^3 - 6x^2 + 12x - 8 = 0$ $[(x - 2)^3]$, $x_0 = 1.7$.
7. $x^4 - 4x^3 + 6x^2 - 4x + 1 = 0$ $[(x - 1)^4]$, $x_0 = 1.3$.
8. $x^4 - 2x^2 + 1 = 0$ $[(x^2 - 1)^2]$, $x_0 = 1.2$ (root $+1$).
9. $x^4 - 8x^2 + 16 = 0$ $[(x^2 - 4)^2]$, $x_0 = 1.8$ (root $+2$).
10. $x^4 + 2x^3 - 3x^2 - 4x + 4 = 0$ $[(x - 1)^2(x + 2)^2]$, $x_0 = 0.7$.
11. For problem 1, also run *ordinary* Newton and tabulate both to contrast linear vs. quadratic convergence.

9 Experiment 7: Secant Method

Objective

To solve $f(x) = 0$ *without* a derivative by replacing the tangent slope with the slope of a secant through the last two iterates.

9.1 Theory

Approximating $f'(x_i) \approx \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}}$ in Newton's formula gives

$$x_{i+1} = \frac{x_{i-1}f_i - x_i f_{i-1}}{f_i - f_{i-1}}, \quad f_i = f(x_i),$$

which needs *two* starting values and *no* bracket. Its order of convergence is ≈ 1.618 (super-linear).

Algorithm 7 Secant Method

Input: f , two initial values x_0, x_1 , ε , N

Output: approximate root x_2

```

1:  $f_0 \leftarrow f(x_0); f_1 \leftarrow f(x_1)$ 
2: for  $n = 1$  to  $N$  do
3:    $x_2 \leftarrow \frac{x_0 f_1 - x_1 f_0}{f_1 - f_0}$ 
4:   if  $|x_2 - x_1| < \varepsilon$  then
5:     return  $x_2$ 
6:   end if
7:    $x_0 \leftarrow x_1; f_0 \leftarrow f_1; x_1 \leftarrow x_2; f_1 \leftarrow f(x_2)$ 
8: end for
9: return  $x_2$ 
```

9.2 Worked Example (full input and output)

Example 7.1 — $f(x) = x^3 - 2x - 5 = 0$

Input. Two seeds $x_{-1} = 2$ ($f = -1$) and $x_0 = 3$ ($f = 16$).

Output.

x_i	f_i
2.05882	-0.39080
2.08126	-0.14720
2.09482	+0.00304
2.09455	≈ 0

$\xi \approx 2.0948$ (x_3 already correct to 3 significant figures).

Example 7.2 — $f(x) = xe^x - 1 = 0$

Input. $x_0 = 0$ ($f = -1$), $x_1 = 1$ ($f = 1.71828$).

Output. $x_2 = 0.36788$, $x_3 = 0.50332$, $x_4 = 0.57861$, $x_5 = 0.56653$, $x_6 = 0.56714 \Rightarrow \xi \approx 0.5671$ (4 d.p.).

Exercise 7 (Secant)

1. $x^3 - 2x - 5 = 0$, $x_0 = 2$, $x_1 = 3$.
2. $xe^x - 1 = 0$, $x_0 = 0$, $x_1 = 1$.
3. $x^3 - x - 1 = 0$, $x_0 = 1$, $x_1 = 2$.
4. $\cos x - xe^x = 0$, $x_0 = 0$, $x_1 = 1$.
5. $x - \cos x = 0$, $x_0 = 0$, $x_1 = 1$.
6. $x^{2.2} - 69 = 0$, $x_0 = 5$, $x_1 = 8$.
7. $x^4 - x - 10 = 0$, $x_0 = 1$, $x_1 = 2$.
8. $e^x - 3x = 0$, $x_0 = 0$, $x_1 = 1$.
9. $\ln x - x + 2 = 0$, $x_0 = 3$, $x_1 = 4$.
10. $x \sin x - 1 = 0$, $x_0 = 1$, $x_1 = 2$.
11. $\tan x - x = 0$, $x_0 = 4.4$, $x_1 = 4.6$.
12. **(Application)** The Colebrook friction factor f obeys $\frac{1}{\sqrt{f}} + 2 \log_{10} \left(\frac{10^{-4}}{3.7} + \frac{2.51}{10^5 \sqrt{f}} \right) = 0$;
use $f_0 = 0.01$, $f_1 = 0.03$.

10 Experiment 8: Ramanujan's Method (Smallest Root)

Objective

To extract the *numerically smallest* root of $f(x) = 0$ directly from its coefficients using Ramanujan's series procedure — no initial guess and no derivative.

10.1 Theory

Write the equation in the standard form $f(x) = 1 - (a_1x + a_2x^2 + a_3x^3 + \dots) = 0$ (divide so the constant term is 1). Expand the reciprocal $1/f(x) = b_1 + b_2x + b_3x^2 + \dots$, where

$$b_1 = 1, \quad b_k = a_1b_{k-1} + a_2b_{k-2} + \dots + a_{k-1}b_1.$$

The *convergents* b_{i-1}/b_i approach, in the limit, the smallest root. Convergence is linear, so many terms may be needed.

Algorithm 8 Ramanujan's Method

Input: coefficients $a_1, a_2, \dots$ of $f(x) = 1 - (a_1x + a_2x^2 + \dots)$, number of terms M , ε

Output: smallest root (last convergent)

```

1:  $b_1 \leftarrow 1$ 
2: for  $k = 2$  to  $M$  do
3:    $b_k \leftarrow \sum_{j=1}^{k-1} a_j b_{k-j}$ 
4: end for
5: for  $i = 2$  to  $M$  do
6:    $c_i \leftarrow b_{i-1}/b_i$  ▷ successive convergents
7:   if  $|c_i - c_{i-1}| < \varepsilon$  then
8:     return  $c_i$ 
9:   end if
10: end for
11: return  $c_M$ 

```

10.2 Worked Example (full input and output)

Example 8.1 — $f(x) = x^3 - 9x^2 + 26x - 24 = 0$ (roots 2, 3, 4)

Input. Divide by -24 : $f(x) = 1 - (\frac{13}{12}x - \frac{3}{8}x^2 + \frac{1}{24}x^3)$, so $a_1 = \frac{13}{12}$, $a_2 = -\frac{3}{8}$, $a_3 = \frac{1}{24}$.

Output.

n	1	2	3	4	5	6	7	8
b_n	1.0000	1.0833	0.7986	0.5006	0.2880	0.1575	0.0835	0.0434
b_{n-1}/b_n	—	0.9231	1.3565	1.5954	1.7384	1.8282	1.8861	1.9242

The convergents climb toward the smallest root $\boxed{\xi = 2}$.

Example 8.2 — $3x - \cos x - 1 = 0$

Input. Using $\cos x = 1 - \frac{x^2}{2!} + \dots$ gives $f = 2[1 - (\frac{3}{2}x + \frac{x^2}{4} - \frac{x^4}{48} + \dots)]$, so $a_1 = \frac{3}{2}$, $a_2 = \frac{1}{4}$, $a_3 = 0$, $a_4 = -\frac{1}{48}, \dots$

Output. Convergents 0.66667, 0.60000, 0.60606, 0.60736, 0.60708, 0.60710, $\dots \Rightarrow \boxed{\xi \approx 0.6071}$ (4 d.p.).

Exercise 8 (Ramanujan's Method)

Find the smallest (in magnitude) root.

1. $x^3 - 9x^2 + 26x - 24 = 0$ (roots 2, 3, 4).
2. $xe^x = 1$ (expand e^x as a series).
3. $3x - \cos x - 1 = 0$.
4. $1 - x + \frac{x^2}{(2!)^2} - \frac{x^3}{(3!)^2} + \cdots = 0$.
5. $x^2 - 5x + 6 = 0$ (roots 2, 3).
6. $x^3 - 6x^2 + 11x - 6 = 0$ (roots 1, 2, 3).
7. $2x^2 - 8x + 3 = 0$.
8. $x^3 - 4x^2 + 5x - 2 = 0$ $[(x-1)^2(x-2)]$.
9. $x^3 - 2x^2 - 5x + 6 = 0$ (roots 1, 3, -2).
10. $6x^3 - 11x^2 + 6x - 1 = 0$ (roots 1, $\frac{1}{2}$, $\frac{1}{3}$).
11. For problem 7, verify your answer against the exact value $\frac{8-\sqrt{40}}{4}$.

11 Experiment 9: Iteration Method for Systems of Nonlinear Equations

Objective

To solve a pair $f(x, y) = 0$, $g(x, y) = 0$ by the fixed-point form $x = F(x, y)$, $y = G(x, y)$ and to verify the two-variable convergence condition.

11.1 Theory

Rewrite the system as $x = F(x, y)$, $y = G(x, y)$ and iterate

$$x_{n+1} = F(x_n, y_n), \quad y_{n+1} = G(x_n, y_n).$$

Convergence to the root (α, β) is guaranteed if, near it,

$$\left| \frac{\partial F}{\partial x} \right| + \left| \frac{\partial F}{\partial y} \right| < 1 \quad \text{and} \quad \left| \frac{\partial G}{\partial x} \right| + \left| \frac{\partial G}{\partial y} \right| < 1.$$

Algorithm 9 Iteration Method for Two Equations

Input: F, G (with the convergence condition), (x_0, y_0) , ε , N

Output: approximate root (x_1, y_1)

```

1: for  $n = 1$  to  $N$  do
2:    $x_1 \leftarrow F(x_0, y_0)$ ;  $y_1 \leftarrow G(x_0, y_0)$ 
3:   if  $|x_1 - x_0| < \varepsilon$  and  $|y_1 - y_0| < \varepsilon$  then
4:     return  $(x_1, y_1)$ 
5:   end if
6:    $x_0 \leftarrow x_1$ ;  $y_0 \leftarrow y_1$ 
7: end for
8: return  $(x_1, y_1)$ 
    
```

11.2 Worked Example (full input and output)

Example 9.1 — $y^2 - 5y + 4 = 0$, $3yx^2 - 10x + 7 = 0$

Input. $x = F = \frac{1}{10}(3yx^2 + 7)$, $y = G = \frac{1}{5}(y^2 + 4)$; $(x_0, y_0) = (0.5, 0.5)$. (At $(0.5, 0.5)$: $|F_x| + |F_y| = 0.225 < 1$ and $|G_x| + |G_y| = 0.2 < 1$.)

Output.

n	x_n	y_n
0	0.5000	0.5000
1	0.7375	0.8500
2	0.8387	0.9445
3	0.8993	0.9784
4	0.9374	0.9914
5	0.9613	0.9966
6	0.9763	0.9986
7	0.9855	0.9994

$(x_n, y_n) \rightarrow \boxed{(1, 1)}$.

Exercise 9 (Iteration for Systems)

Rewrite each system as $x = F(x, y)$, $y = G(x, y)$ so that the convergence condition holds, then iterate from the given start.

1. $y^2 - 5y + 4 = 0$, $3yx^2 - 10x + 7 = 0$; $(0.5, 0.5)$.
2. $x^2 + y^2 - 4 = 0$, $xy - 1 = 0$; $(1.9, 0.5)$.
3. $x^2 - 2x - y + 0.5 = 0$, $x^2 + 4y^2 - 4 = 0$; $(0, 1)$.
4. $x^2 + y^2 - 1 = 0$, $y - x^2 = 0$; $(0.8, 0.6)$.
5. $x = \frac{1}{10}(x^2 + y^2 + 8)$, $y = \frac{1}{10}(xy^2 + x + 8)$; $(0, 0)$.
6. $\sin x + y^2 - 1 = 0$, $x + \cos y - 1.2 = 0$; $(0.5, 0.5)$.
7. $x = 0.5 + 0.2 \sin(x + y)$, $y = 1 + 0.1 \cos(x - y)$; $(0.5, 1)$.
8. $2x^2 - xy - 5x + 1 = 0$, $x + 3 \log_{10} x - y^2 = 0$; $(3.5, 2.2)$.
9. $x^2 + y - 11 = 0$, $x + y^2 - 7 = 0$; $(3.5, 1.8)$.
10. $x^2 - y - 1 = 0$, $x - y^2 + 1 = 0$; $(1.5, 1.5)$.
11. For each system, evaluate the partial derivatives at the start and state whether the convergence condition is satisfied.

12 Experiment 10: Newton–Raphson Method for Systems

Objective

To solve $f(x, y) = 0$, $g(x, y) = 0$ by Newton's method in two variables, linearising with the Jacobian and updating with Cramer's rule.

12.1 Theory

With (x_0, y_0) a guess and $(x_0 + h, y_0 + k)$ the root, Taylor's expansion to first order gives the linear system

$$hf_x + kf_y = -f_0, \quad hg_x + kg_y = -g_0,$$

where the partials are evaluated at (x_0, y_0) . With the Jacobian $D = \begin{vmatrix} f_x & f_y \\ g_x & g_y \end{vmatrix} \neq 0$, Cramer's rule yields

$$h = \frac{1}{D} \begin{vmatrix} -f_0 & f_y \\ -g_0 & g_y \end{vmatrix}, \quad k = \frac{1}{D} \begin{vmatrix} f_x & -f_0 \\ g_x & -g_0 \end{vmatrix}, \quad x_1 = x_0 + h, \quad y_1 = y_0 + k.$$

Convergence is second order.

Algorithm 10 Newton–Raphson Method for Two Equations

Input: f, g and partials f_x, f_y, g_x, g_y , (x_0, y_0) , ε , N

Output: approximate root (x_0, y_0)

```

1: for  $n = 1$  to  $N$  do
2:   evaluate  $f_0, g_0, f_x, f_y, g_x, g_y$  at  $(x_0, y_0)$ 
3:    $D \leftarrow f_x g_y - g_x f_y$  ▷ Jacobian; require  $D \neq 0$ 
4:    $h \leftarrow (-f_0 g_y + g_0 f_y) / D$ 
5:    $k \leftarrow (-g_0 f_x + f_0 g_x) / D$ 
6:    $x_0 \leftarrow x_0 + h$ ;  $y_0 \leftarrow y_0 + k$ 
7:   if  $|h| < \varepsilon$  and  $|k| < \varepsilon$  then
8:     return  $(x_0, y_0)$ 
9:   end if
10: end for
11: return  $(x_0, y_0)$ 
```

12.2 Worked Example (full input and output)

Example 10.1 — $f = 3yx^2 - 10x + 7$, $g = y^2 - 5y + 4$

Input. $f_x = 6yx - 10$, $f_y = 3x^2$, $g_x = 0$, $g_y = 2y - 5$; $(x_0, y_0) = (0.5, 0.5)$.

Output. First step: $f_0 = 2.375$, $g_0 = 1.75$, $D = \begin{vmatrix} -8.5 & 0.75 \\ 0 & -4 \end{vmatrix} = 34$, $h = 0.3180$, $k = 0.4375$.

n	x_n	y_n
0	0.50000	0.50000
1	0.81801	0.93750
2	0.97079	0.99875
3	0.99934	1.00000

→ $(1, 1)$ in ~ 3 steps (iteration needed 7+).

Example 10.2 — $x^2 + y^2 = 1, \quad y = x^2$ (circle $\cap$ parabola)

Input. $f = x^2 + y^2 - 1, \quad g = y - x^2; \quad f_x = 2x, \quad f_y = 2y, \quad g_x = -2x, \quad g_y = 1; \quad (x_0, y_0) = (0.7071, 0.7071).$

Output. $(0.7071, 0.7071) \rightarrow (0.7929, 0.6213) \rightarrow (0.7862, 0.6180) \Rightarrow \boxed{(0.7861, 0.6180)}.$

Exercise 10 (Newton–Raphson for Systems)

List the four partials, then iterate from the given start to 4 d.p.

1. $3yx^2 - 10x + 7 = 0, \quad y^2 - 5y + 4 = 0; \quad (0.5, 0.5).$
2. $x^2 + y^2 - 1 = 0, \quad y - x^2 = 0; \quad (0.8, 0.6).$
3. $\sin x - y + 0.9793 = 0, \quad \cos y - x + 0.6703 = 0; \quad (0.5, 1.5).$
4. $x^2 + xy + y^2 - 7 = 0, \quad x^3 + y^3 - 9 = 0; \quad (1.5, 0.5).$
5. $x^2 - 2x - y + 0.5 = 0, \quad x^2 + 4y^2 - 4 = 0; \quad (2, 0.25).$
6. $x^2 + y^2 - 4 = 0, \quad xy - 1 = 0; \quad (1.9, 0.6).$
7. $e^x - y = 0, \quad xy - e^x = 0; \quad (0.8, 1.5).$
8. $x^2 + y - 11 = 0, \quad x + y^2 - 7 = 0; \quad (3.5, 1.8).$
9. $x^2 - y - 1 = 0, \quad x - y^2 + 1 = 0; \quad (1.5, 1.5).$
10. **(Application, 2-link robot arm, $L_1 = L_2 = 1$)** $\cos \theta_1 + \cos(\theta_1 + \theta_2) = 1.2, \quad \sin \theta_1 + \sin(\theta_1 + \theta_2) = 0.8; \quad (0, 1.5).$
11. For problem 4, verify that $(2, 1)$ is an exact root and discuss which root your iteration converges to.

Appendix: Quick Comparison of the Methods

Method	Update / formula	Order	Needs
Bisection	$x = (a + b)/2$	Linear	bracket $f(a)f(b) < 0$
False position	$x = \frac{af(b) - bf(a)}{f(b) - f(a)}$	Linear	bracket
Iteration	$x_{n+1} = \varphi(x_n)$	Linear	$ \varphi' < 1$
Aitken Δ^2	$\hat{x} = x_{i+1} - \frac{(\Delta x_i)^2}{\Delta^2 x_{i-1}}$	accelerated	3 iterates
Newton–Raphson	$x_{n+1} = x_n - \frac{f}{f'}$	Quadratic	f, f' , one x_0
Generalized Newton	$x_{n+1} = x_n - p \frac{f}{f'}$	Quadratic	multiplicity p
Secant	$x_{i+1} = \frac{x_{i-1}f_i - x_i f_{i-1}}{f_i - f_{i-1}}$	≈ 1.618	two seeds
Ramanujan	$b_{i-1}/b_i \rightarrow$ smallest root	Linear	coefficients
Iteration (systems)	$x_{n+1} = F, y_{n+1} = G$	Linear	$ F_x + F_y < 1$
Newton (systems)	solve $J[h \ k]^T = -[f \ g]^T$	Quadratic	Jacobian $D \neq 0$

Reference: S.S. Sastry, *Introductory Methods of Numerical Analysis*, 5th ed., PHI Learning — § 2.2–2.7, 2.12.